

Continuous Hairpin Winding Technology for Electric Machines Enabling Net Zero Transportation: A Comprehensive Review

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ABSTRACT Featured by low power losses and strong heat dissipation capability, hairpin winding has been widely accepted as a key enabler of boosting performance for traction/propulsion electrical machines in automotive and aerospace sectors. Continuous hairpin winding (CHW), distinguished by its further improvement brought to machines' key performance indicators, being power density, efficiency, and reliability, is emerging as a more promising winding solution. This paper will provide a critical technology review on CHW, with focus on highlighting its different features compared with those of other typical winding types, manufacturing/assembly process, layout design rules, as well as current and future development trends. Based on illustration of the unique winding process, the so-called "radial shift" feature that inherently exists in CHW will be introduced. The key elements in winding layout development, including transposition, terminal and jumper connections, parallel branch number, will be summarised for CHW. Moreover, new technology bricks and research ideas that strive to tackle manufacturing challenges, enhance design flexibility, as well as improve overall performance have been highlighted. Finally, the paper concludes by proposing future research directions, with the vision of increasing its Technology Readiness Level (TRL) for future net-zero transportation.

INDEX TERMS Hairpin winding, net zero transportation, continuous hairpin winding, electrical machines.

NOMENCLATURE

N_s	Number of stator slots.
N_L	Total layer number of conductors in a slot.
p	Number of poles.
m	Number of phases.
q	Slot number per pole per phase.
N_{aa}	Parallel branch number per phase.
N_{wire}	Number of inlet wires wound in CHW.
$N_{wire/phase}$	Number of wires per phase in CHW.
$N_{tp-type}$	Number of transposition types of a wire.

I. INTRODUCTION

Electrical conductors have played a pivotal role in power transmission since the discovery of conductivity in 1729, following by practical applications of transmitting electrical signals and energy since 1744. Faraday's discovery of electromagnetic induction in 1831 then paved the way for invention of electrical machines (generators, motors, and transformers), in which conductors were wound as coils and served as central pivot of electromagnetic energy conversion [1], [2].

Winding technology—the art and science of shaping and arranging coils in electrical machines—has been evolving

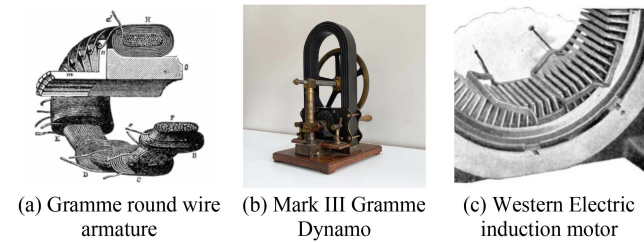


FIGURE 1. Early forms of the coil/armature with: (a) Round wire, (b) formed ring and (c) preformed flat wire, reproduced from [5].

TABLE 1. Early Stage Milestones of Winding Technology

Year	Milestone
1869	Stranded round-wire armature in DC generators [5]
1903	First flat-wire (“hairpin”) coil patented [6]
1906	Introduction of enamel/asbestos insulation on wires in Germany
1911	Roebel’s transposed-bar concept to avoid eddy currents [12]
1927	Vincent G. Apple’s patents on flat armature windings [7][8][9]
1955	DuPont develops polyimide polymers for magnet wire coating
1973	IEC 60317 Standard published for enamelled round wires
1980s	PEEK polymer coating for high temperature/voltage rating

with the aim of performance improvement, cost reduction, and manufacturing automation [3], [4]. At early stage, the main key performance indicator (KPI) of winding, being reducing DC Joule losses, was embodied as fill factor, i.e., ratio of pure conductor area to allocated winding area. The need for easier production and assembly led to the initial use of round wires in stranded winding armature of DC generators, as shown in Fig. 1(a) and 1(b) [5]. The quest for higher fill factor led to the first U-shaped flat conductor solution [6], which was further improved by Vincent. G. Apple [7], [8], [9]. This kind of preformed pattern of flat wire then became known as “hairpin winding” [10].

In 20th century, the manufacturing differences between round and flat wire deviated their application scenarios in electric machines. Round wires were typically wound as windings directly into the iron core, adapting to various slot shapes through multiple turns, featuring high design flexibility and relatively low production costs. Consequently, round wires have been widely used in small to middle power level machines [11]. In contrast, rectangular wires need to be pre-shaped to align with specific slot sizes, which involve extensive insertion and welding processes, leading to reduced flexibility and increased capital costs. Meanwhile, the improved fill factor and maintainability of rectangular wires make them suitable for specialized large machines with small-scale production, such as MW-level hydro generators with Roebel bars [12], [13]. Table 1 listed key milestones of winding technology prior to 1980s.

Over the last decades, KPIs of electric machine windings became more standardised and were extended to multi-physics domains [14]. The ever-more demanding and newly

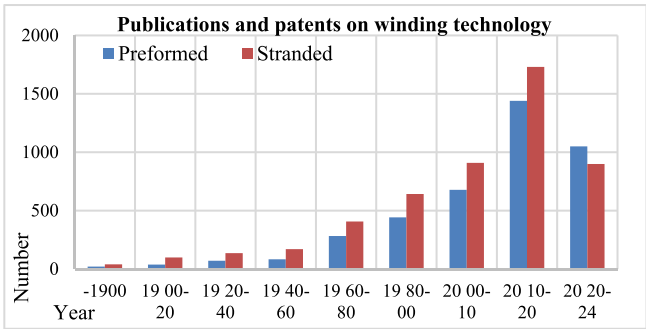


FIGURE 2. Trends in publications (papers and patents) related to preformed windings using flat wire and stranded windings using round wire. Source: Google scholar. Flat wire search terms: (“flat wire” OR “flat conductor” OR “rectangular wire”) AND (“electrical machine” OR “electric motor”). Round wire search terms: (“round wire” OR “round conductor” OR “random wire”) and (“electrical machine” OR “electric motor”).

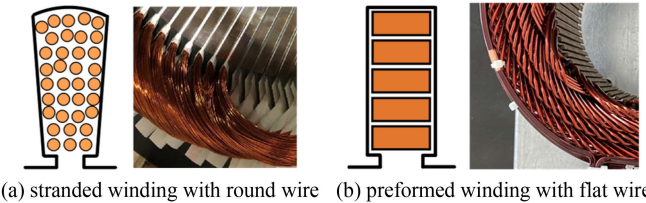


FIGURE 3. Stranded winding and preformed windings, adapted from [22].

emerged application scenarios are pushing for higher conductivity, temperature rating, insulation reliability and mechanical strength of windings. In transportation sector, this has led to further development of both stranded winding (compressed wire [15], Litz wire [16]), and preformed winding (hairpin [17], and edgewise winding [18]). Fig. 2 shows the growth of publications in both winding types, which indicate a notable shift in research interests towards flat wire winding within the last 5 years.

An inherent advantage of flat wire winding is its high slot fill factor (Fig. 3) and much reduced DC Joule losses. Through careful wire size selection as well as connection layout design, flat wire winding can also be advantageous in AC Joule loss mitigation for high-speed applications [17]. Moreover, with significantly improved consistency of conductor placement both in and out of stator slots, intense cooling techniques can be applied, allowing for significantly higher current density / loading [19]. Preformed flat wire also offers better insulation reliability than stranded round wire, due to its uniform geometry and parallel conductor surfaces, which mitigates local electric field peaks and reduces the risk of partial discharge [20]. Its highly automated manufacturing process further ensures consistent thermal and mechanical performance by minimizing stress/thermal concentrations, which further improves the insulation performance [21].

Fig. 4 compares qualitatively the leading KPIs of preformed flat wires and stranded round wires used as electric machine windings, with more details to be discussed in the

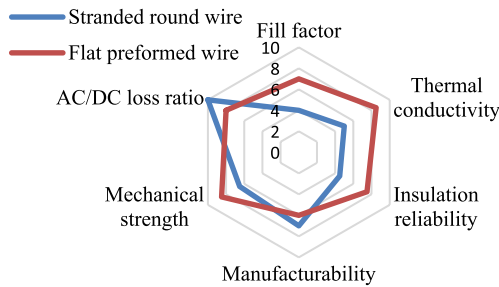


FIGURE 4. Comparison between stranded round and preformed flat wires.

following sections. Nowadays, new solutions are being actively investigated to enhance flexibility and reduce cost of preformed flat wire windings, including multi-layer designs for high frequency applications [17], [23], further improved cooling system [24], and simplified manufacturing process for mass-production [25], [26], [27], [28]. For high-speed, high-power applications, multi-layer hairpin windings (as shown in Fig. 3(b)) have been distinguished and widely applied in electric vehicle (EV) sector, as a key enabler of boosting performance of EV traction motors [29], [30], [31], [32]. Moreover, along with improved maturity, hairpin winding technology is being extended from automotive to aerospace application sector [33], [34]. Pushed by the global “Net Zero” ambition, roadmaps of EV traction and aerospace propulsion are increasingly defined by stringent demands on power density, reliability, and efficiency [35], [36]. This has prompted the adoption of a more prestigious hairpin winding, known as continuous hairpin or endless flat winding [37].

This paper will provide a critical technology review and analysis on continuous hairpin winding (CHW), with a focus on its current and future applications in electrified transportation sector. Section II is devoted to comparing CHW with flat wire based conventional hairpin windings in terms of general KPIs (fill factor, AC/DC losses, torque quality, efficiency) through literature review and analysis. Section III provides a detailed introduction in typical manufacturing process and potential alternatives for CHW, in comparison with that for typical segmented hairpin windings. Section IV summarizes the special layout design rules for CHW with manufacturing and assembly limitations taken into account. Finally, the current challenges and critical issues related to the large-scale application of CHW in transportation will be discussed in Section V, followed by highlights of the latest advances and vision of future directions for CHW research and development.

II. TRANSITION FROM HAIRPIN TO CONTINUOUS HAIRPIN

Traction and automotive machines originated as medium- and small-frame electric motors employing stranded round conductors to form windings. Although this winding topology offered low capital expenditure and mature manufacturing process, it is subject to modest slot fill factors and restricted

heat removal capability. As power-density requirements tightened at the end of the 20th century, industry increasingly sought winding solutions that could exploit the full cross-section of the stator slot.

A breakthrough came in 1999 when Denso Corp. introduced the flat-wire hairpin winding for automotive [38]. The segmented winding imbedded more copper into the slot and provided a direct heat path to the laminations. Meanwhile, this also brought massive end winding welding joints which lead to costly CNC bending equipment, a succession of stripping-, aligning-, twisting- and laser-welding steps for every segment, longer cycle times and a higher likelihood of insulation damage. Subsequent patents therefore focused on trimming the segment count and simplifying welding operations [39], [40]. Remy’s patents [41], [42] then emphasized the importance of transposition for balancing multiple parallel paths to avoid circulating currents and losses.

Along with ambitious performance roadmaps of EV/aircraft powertrains [35], [36], traction/propulsion motor solutions are being pushed to feature higher speed and fundamental operating frequency [43]. To suffice this trend and mitigate AC winding Joule losses, designers progressively reduced individual conductor dimension by increasing both stator slot and layer number. The direct side-effect was an explosion in the quantity of welding points, increased end winding mass/volume, and a higher risk of insulation failure [44].

The continuous hairpin winding (CHW)—developed in parallel with conventional hairpins and already adopted in automotive electrical machines [45], [46], features a single rectangular conductor that is twisted continuously to create one entire phase branch. This eliminated most welding points per branch compared with conventional hairpin windings. In [47], Mitsubishi’s CHW design was claimed to have lowered the risk of insulation degradation, improved cooling quality, and reduced winding loss. They later combined the CHW with flat lamination bending technique, eliminated the need to shoe-horn windings into the cylindrical stator [48]. The fundamental winding pattern which configured conductors’ layout between stator slots and layers were mapped in [49], and a full manufacturing route was detailed in [50].

To further improve the compactness of end winding, Denso Corp. patented a stepwise CHW (Fig. 5), in which the terraced end turns shortened axial overhang and reduced DC winding resistance. The same geometry also eliminated inter-coil interference [51], and promoted better coolant circulation [52]. Elemtotec complemented this work with guiding fixtures that steer each sinusoidal conductor through the lamination stack [53]. Subsequent studies showed that pre-interweaving multiple wires prior to insertion could yield more compact and mechanically robust assembly [54]. Over the past decade, the rapid advancement of electrification in automotive sector has spurred extensive investigations into both hairpin and CHW technologies. Fig. 6 shows the end windings of hairpin and CHW from two state-of-the-art EV products.

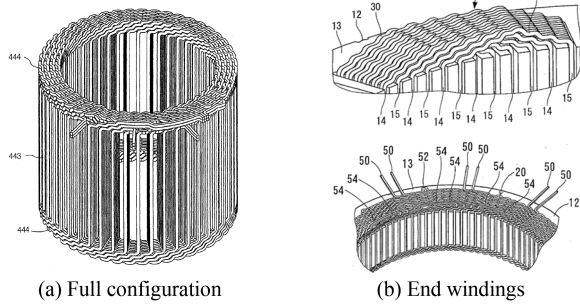


FIGURE 5. Stepwise continuous hairpin, reproduced from [51], [52].

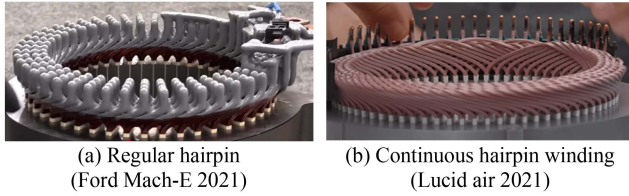


FIGURE 6. Typical hairpin and continuous hairpin winding products.

TABLE 2. Summary of Key Studies on CHW Over the Past Decade in Comparison With Other Winding Types

Paper	CHW Type	Counterpart	End length	DC loss	Other KPIs (Efficiency, torque density, magnet usage.)
[55]	Lap	Round DW	-14%	-20%	Efficiency: +3.7% @ 1krpm +1.1% @ 9krpm
[56]	Lap	Hairpin	-2.2%	-5.6%	Welding points: -75%
[57]	CHW	Hairpin	-	-	Winding temperature: -7°C WLTP Efficiency: +5%
[58]	CHW	Hairpin	-10mm	-9%	Efficiency: +1.2% @ peak +0.3% @ rated
[59]	CHW	Round CW	+92%	+39%	Torque density: +35% Magnet weight: -35% Efficiency: +1%
[60]	CHW	Hairpin	-	-	Process steps: -50%
[61]	CHW	Hairpin	-	-	WLTP Efficiency: +0.63% Iron loss: -11.3% Copper loss: +10.7%
[62]	CHW	Hairpin	-10mm	-6.6%	Copper loss: -3.9% Average efficiency: +0.1% Power density: +1.7%

DW: distributed winding, CW: concentrated winding.

From academic perspective, a few recent literatures have underscored advantages of CHW in electromagnetic and manufacturing aspects through quantified comparison with counterpart winding options (see Table 2). Ishigami et al. [55], [56] first proposed a continuous diamond-lap configuration that eliminates 75% of the weld joints compared to conventional hairpin, gaining the benefit of 6% reduction in DC resistance. In [57], Schaeffler GmbH found that CHW enables stator design with more slots and smaller conductors, improving heat dissipation and reducing high-frequency AC losses. This led to lower winding temperatures and 5%

TABLE 3. Features and Performances of Flat Wire Windings

Flat wire windings	Lap	I-Pin	Hairpin		CHW*	
			Regular	X-pin	Sword	Flat
Welding-less	+	--	-	-	++	++
Mecha. strength	0	-	-	--	+	++
Insulation reliability	+	-	0	-	+	++
Fill factor	0	++	+	+	+	+
Max. possible slots	0	-	-	-	+	+
Max. possible layers	0	--	--	-	+	++
End compactness	-	--	-	+	++	++
Slot opening type	open	any	any	any	open	open
Assembly direction	radial	axial	axial	axial	radial	radial

Rating scale: ++: Excellent, +: Good, 0: Neutral, -: Poor, --: Very Poor

higher WLTP efficiency compared to hairpin winding. Song et al. [58] observed that the shortened end windings of CHWs enable a 9% reduction in conductor usage and 0.3%-1.2% improvement in efficiency, albeit with increased cogging torque due to the open slot structure. Husain et al. [59] validated this efficiency improvement in low-torque regions and emphasized mass reduction and associated cost savings. Kampker et al. [60] noted that eliminating welding steps can streamline the manufacturing process, though it may introduce operational complexities. Marjuan et al. [61] examined drive-cycle performance and harmonic distortion, respectively, reporting a 0.63% increase in WLTP efficiency and enhanced demagnetization resistance. However, the same group also highlighted the trade-off of elevated back-EMF harmonics due to the wide slot openings. In [62], a case study demonstrated that CHW can lower the insulation risk by eliminating welding, reduce end winding length, and enhance power density, making it a viable candidate for aerospace MW-class electrical machines. Collectively, these studies confirm the benefits of CHW in minimizing end winding length and resistance, increasing possible slot/layer numbers, improving efficiency, with challenges in cogging torque and torque ripple brought by the open-slot structure.

Table 3 benchmarks six commonly used flat-conductor windings in EV traction motors—diamond-lap, I-Pin, regular hairpin, X-pin (A variant of hairpin with reduced height of welding joints through advanced welding), and continuous hairpin—against qualitative metrics derived from their insertion type (radial vs. axial) and the companion slot architecture (open vs. semi-closed) [63]. Based on this, two major points can be concluded as below:

A. HIGH FREQUENCY AC JOULE LOSS

CHW features the same AC loss mechanisms compared with its counterparts. Fig. 7 illustrates typical AC effects of flat conductors (fed with AC current of 1000 Hz as example). At conductor level, AC loss is induced by skin and proximity effect under the influence of in-slot magnetic field. At bundle level, additional AC loss could be generated when different

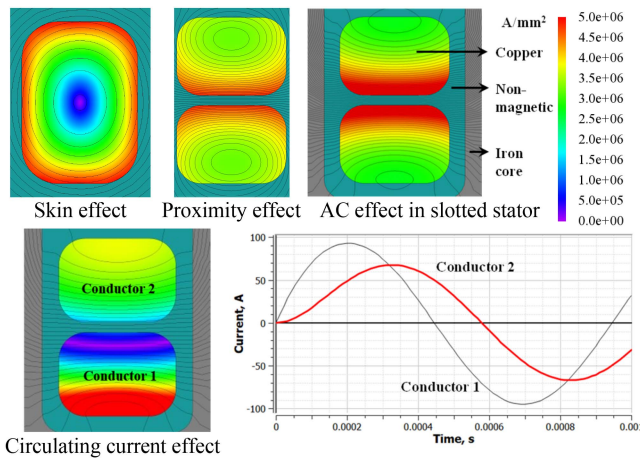


FIGURE 7. Illustration of AC loss effects in hairpin winding and CHW.

parallel branches feature asymmetric impedance. This is embodied as circulating current between branches, and can be effectively mitigated by posing specific layout design rule mentioned as transposition [41], [42] (further discussion in Section V). It should be noted that both conventional hairpin winding and CHW might be subject to all the AC effects above. On one hand, the fact that CHW can be designed with higher slot number and winding layer number is beneficial to the mitigation of conductor level AC loss. While on the other hand, it will be pointed out in Section V that CHW might be more easily subject to additional circulating current induced AC loss due to its more problematic transposition design.

Another factor to mention is that radial-insertion windings—Lap and CHW—require an open-slot stator. This increases slot-leakage reactance and typically elevates AC loss of conductors near the slot opening area. Magnetic or hybrid wedges can reduce the slot leakage flux, but they introduce extra cost and assembly steps [64]. These open-slot penalties diminish as slot number rises, making CHW particularly well-suited to the high slot number, high frequency machines targeted for next-generation traction propulsion applications. Overall, CHW features the potential of further reducing AC loss compared with other flat winding options, while based on limited literature on this topic, there is no absolute conclusion that CHW is already the most advantageous solution in AC loss mitigation.

B. DEPENDENCE ON WELDING AND ITS INFLUENCE

The number of welds falls progressively from I-Pin and conventional hairpin, through diamond-lap variants, to its lowest in CHW. Each reduction shortens cycle time, lowers capital expenditure, and removes the thermal-mechanical hotspots that can trigger partial discharge. All flat-wire topologies already exceed stranded coils in dielectric endurance, yet CHW—by eliminating every branch weld—delivers the most uniform and reliable insulation system for high-voltage,

inverter-fed motor drives. Apart from compromising insulation integrity and lengthening the process chain, welding also enlarges the end winding envelope—both radially and circumferentially—thereby restricting stator designs with many slots or radial layers. Under present manufacturing practice, welded hairpin solutions struggle to exceed 10 layers and 96 slots. In contrast, established CHW lines from GROB/Schaeffler already support 16-layer conductors in stators with more than 120 slots [65]. This scalability is pivotal for further downsizing flat conductors, curbing conductor level AC losses, increasing exposed surface for effective cooling, and therefore meeting the higher frequency, higher power density requirements of next-generation traction machines.

Despite these encouraging findings, the current body of work on CHW still exhibits a pronounced knowledge gap. In academia, there is a lack of comprehensive synthesis on manufacturing techniques, as well as universally applicable design guidelines, to thoroughly summarise CHW's state-of-the-art technologies. Particularly, there are extensive research elements in the winding layout theory of CHW, including critical areas of parallel branch and transposition design, which are yet to be summarised. This absence of a holistic framework restricts the broader application and further optimization of the promising CHW technology in future traction/propulsion motors.

III. MANUFACTURING PROCESS OF CHW

A. OVERALL MANUFACTURING PROCESS

A typical segmented hairpin winding involves several welding-related process steps, as outlined in Table 4. These mechanical and thermal operations not only complicate the manufacturing process but also elevate the risk of insulation failure. In contrast, CHW features significantly reduced steps and hence more streamlined and efficient manufacturing process, as presented in Fig. 8. The winding process is critical in CHW manufacturing [65], in which wires guided by the wire nozzles are to be continuously wound on a winding sword to form a flat wire mat with two layers.

The transposition of wires, which will be introduced in detail in Section III-E, is also accomplished during this winding step. Subsequently, the wire mat is gradually fed from one end of the winding sword and pressed into a rotating circular positioning gripper to form a wound wire roll. The wire roll is then expanded outward and inserted into the open slot stator with insulation liner. This will be followed by the bus bar welding for terminal and neutral point connections. Afterwards, the assembly of CHW is essentially completed and the whole stator can be impregnated [66]. The manufacturing steps above impose specific design constraints on CHW, distinct from regular hairpin windings. For example, the way the wire mat is rolled up dictates the connection sequence of phase conductors, and the number of inlet wires puts a limit on the maximum number of parallel branches of phases. It should be noted that the AC winding theory needs to be considered and followed in this manufacturing process, which introduces

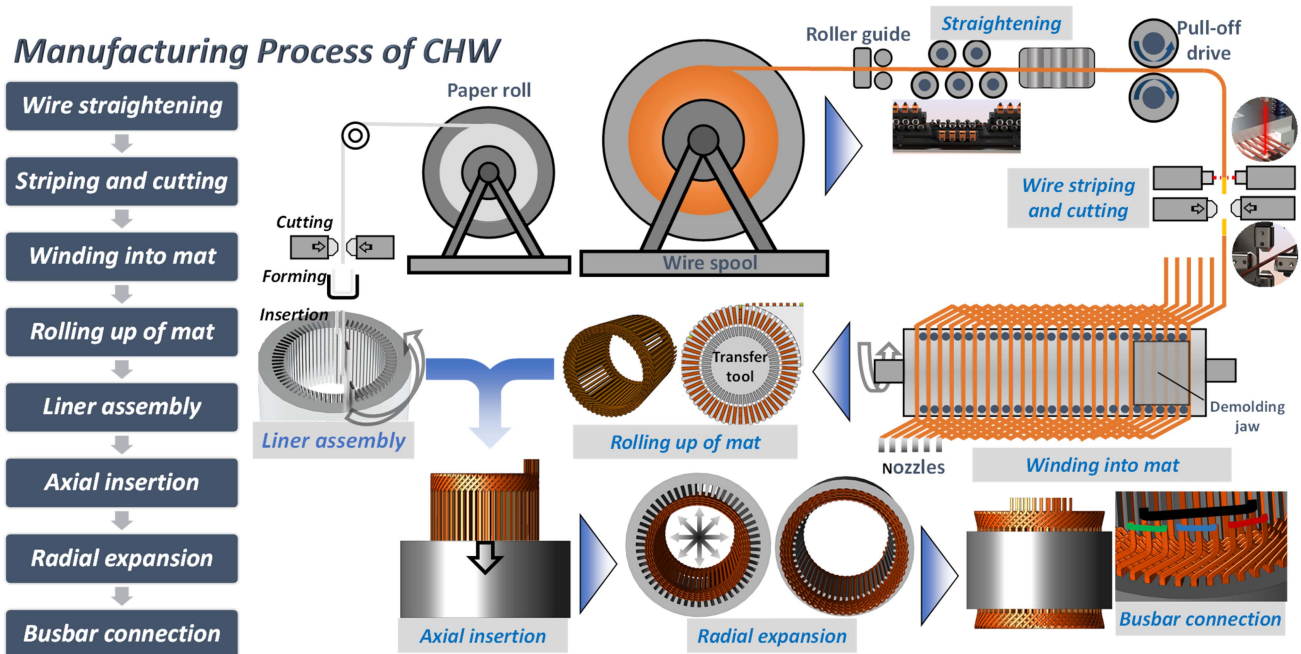


FIGURE 8. A typical manufacturing process of continuous hairpin winding [65], [66].

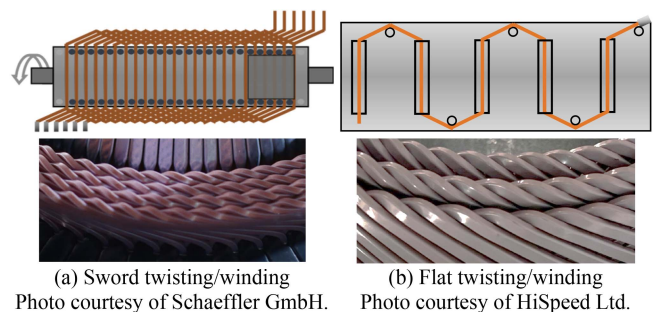
the important concept of transposition [17]. For CHW, transposition consisting of layer and phasor change jumpers, is a key factor that influences manufacturing complexity.

Apart from the typical process shown in Fig. 8, there are alternative approaches at each step of the process. The subsequent sub-sections will introduce various options concerning twisting/winding methods, winding direction, winding patterns, as well as the all-important transposition. This detailed review aims to comprehensively unveil how the current state of art manufacturing technology impacts the guidelines for CHW development.

B. TWISTING/WINDING METHODS OF CHW

There are two primary twisting methods, differentiated by the type of winding platform used. The first method is based on a winding sword, which rotates to wound the wires. The wire mat is then removed from the winding sword, collected, and compressed onto a buffer [67]. The second method uses a flat winding board, where wires are bent independently or in groups with forming tools [68]. These bent wires are then stacked or alternately plugged to form the entire winding mat [69], [70]. As shown in Fig. 9(a) and (b), the shape of wire head differs between these two methods.

Due to processing methods, the above winding and mat forming methods also lead to different layouts, i.e., electrical connection maps of “in-slot” conductors, especially in transposition arrangements. In sword winding method, two sets of wires are interwoven through the sword, allowing them to switch layer positions at each end connection.



(a) Sword twisting/winding
Photo courtesy of Schaeffler GmbH.

(b) Flat twisting/winding
Photo courtesy of HiSpeed Ltd.

FIGURE 9. Two wire mat manufacturing methods, adapted from [60].

In flat winding process, the layer change for transposition is not fulfilled [71], however can be achieved at the following steps of assembly process, depending on various options proposed. Denso Corp. first discussed the interlacing and insertion steps to achieve layer change at specific positions [72]. Subsequent patents demonstrate that plugging tools and automated assembly processes can be applied to achieve layer change in other ways [73], [74], [75]. Typically, flat twisting demands lower processing forces on the wire and smaller end winding height [76], but it involves more sub-processing steps to achieve layer transposition. The subsequent discussion will focus on the sword CHW.

C. WINDING DIRECTIONS OF CHW

In addition to the twisting methods, there are also two common options regarding the winding direction, firstly discussed in [77]. The direction, referring to the movement of the wire

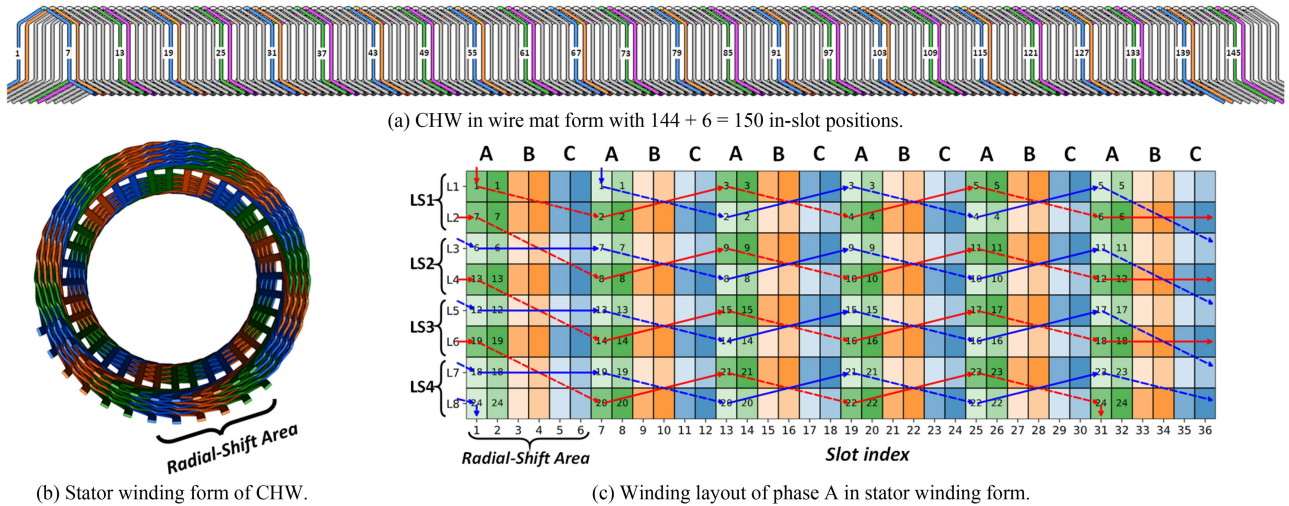


FIGURE 10. A contrast of winding layout of the typical CHW under the wire mat form and stator winding form.

TABLE 4. Process Steps of Typical Hairpin Winding

Steps	WR*	Description / ● Influences	Effect
Wire straightening		Straightened to remove curvature ● Multi-sets of straightening tools	+ Cycle time
Wire cutting	●	Cut wire to the required length ● Deformation on wire cross-section ● Cracks/roughness (heat concentration)	+ Copper loss - Weld quality
Wire laser skinning	●	Insulation stripped by pulsed laser ● Thermal stress on insulation	+ Energy use - Insulation
Pin shape preforming	●	Bending through CNC, Swivel or die ● Machinery/equipment investments ● Spring-back compensation of wire	- Fill factor - Insulation
Pin pre-assembly & insertion	●	Axial insertion of pin baskets into stator ● Scratches on insulation ● Stress/deformation on wire	- Fill factor - Insulation
Separating wire ends	●	Radial separate wire ends for twisting ● Increased winding end height/volume	+ Cycle time + End volume
Twisting wire ends	●	Wire ends twisted tangentially with rotary tool and aligned for welding ● Scratches/ mechanical stress on coating	+ Time / Cost - Insulation
Wire ends final cutting	●	Height alignment of ends for welding	+ Cycle time
Laser welding	●	Wire ends melted connected by laser and electrically connected ● Risk of failure/insufficient connection ● Thermal stress on insulation	+ Energy use + Copper loss - Insulation
Post-weld insulating	●	Weld side re-coated via epoxy powder ● Risk of failure/insufficient insulation ● Increased weight of winding	+ Time / Cost + Mass - Insulation

*WR: Welding-related steps.

nozzle during the winding process, determines the relative positions of the inlet and outlet terminals, as well as the total length of each wire. One option corresponds to the wire nozzle moving consistently forward, named as the forward winding (FW). In this method, the start and end points of each wire are at opposite ends of the wire mat, corresponding to the first and last layers of the stator winding, as shown in Fig. 11(a). The other option is to halve the number of inlet wires, as

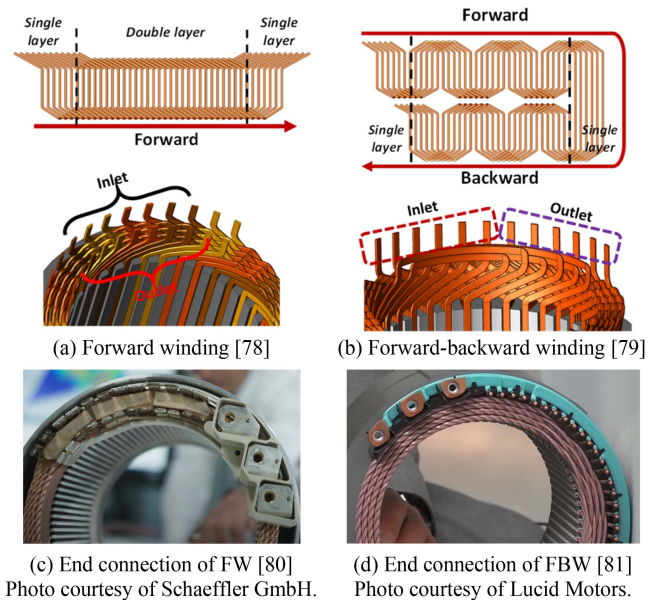


FIGURE 11. Winding direction: forward and forward-backward winding.

shown in Fig. 11(b), so that the wire initially occupies only half of the positions. When the nozzle reaches the end of the wire mat, it reverses direction to twist the wire and occupy the remaining positions. This method is therefore termed as forward-backward twist (FBW), where each wire's start and end points are on the same side of the mat, or the same layer of the stator. As can be seen in Fig. 11, the bus bar connections are also different in corresponding to the positions of wire terminals.

D. RADIAL SHIFT EFFECT AND MITIGATION

The nature of CHW manufacturing process corresponds to an end shift effect, leading to an unbalanced layout. Seen from

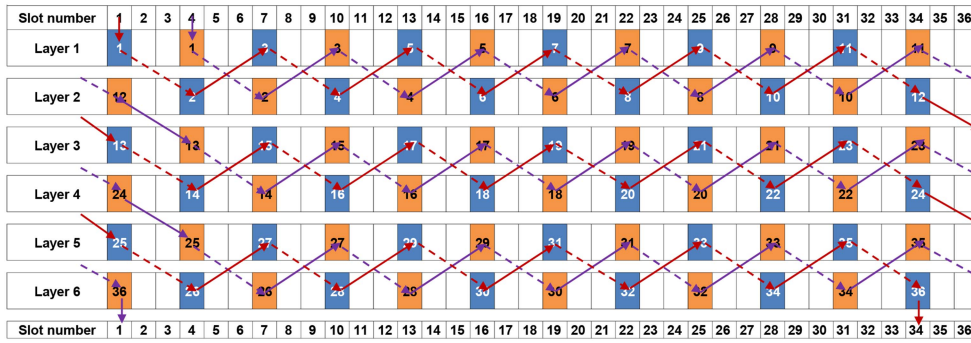


FIGURE 12. CHW pattern without the radial shift effect, adapted from [49].

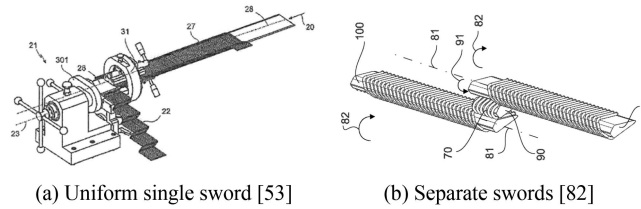


FIGURE 13. The sword configurations of CHW for (a) uniform single sword for the radial-shift pattern, reproduced from [53], (b) Separate swords for the non-radial-shift pattern, reproduced from [82].

the flat wire mat shown in Fig. 10(a), it can be observed that the majority of the “in-slot” conductors are overlapped, i.e., featuring “two layers”. Meanwhile, at each end of the mat, there are 6 ($m \times q$) positions that are with single layer of conductor. During the rolling of the flat wire mat along circumferential direction, when the mat completes a full circle and returns to the end positions, the conductors at this point shifts radially upwards to fill the missing layer of conductor in the first single layer region. This causes the radial position of the wires in this end area to differ from major areas, named as radial shift effect.

Fig. 10(b) shows the in-slot positions of wires under a winding layout with ($q-1, p-12, m-3, N_L-8$). The radial-shift area (RSA) consists of first 6 slots of stator, in which layer positions of wires are different from the major area. Consequently, blue wire takes an additional layer 2, 4, and 6 from the green wire while gives layer 3, 5, 7 back. As shown in Fig. 10(c), this will then result in the unbalanced distribution of wires at the layer level.

Many researchers have noted that this radial shift effect, with its uneven conductor position distribution, leads to additional asymmetry between parallel branches and additional copper loss. The non-radial-shift pattern has been proposed in [49] to mitigate the effect, as shown in Fig. 12. To implement this pattern and correct the displacement caused by the radial shift, the intertwining of the two wires needs to be paused. Achieving this ‘pause’ requires reversing the winding direction at the radial shift area [82], [83]. One possible method is illustrated in Fig. 13(b), where the wires are wound around

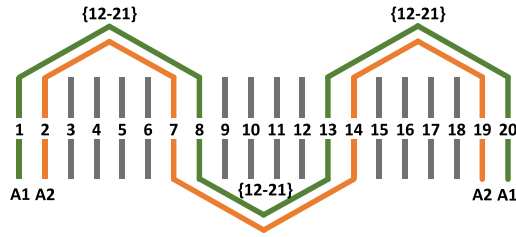
the second and third set of swords to change the winding direction. Conversely, the default radial shift winding pattern only necessitates a uniform winding sword [84], as depicted in Fig. 13(a).

E. TRANSPOSITIONS OF CHW

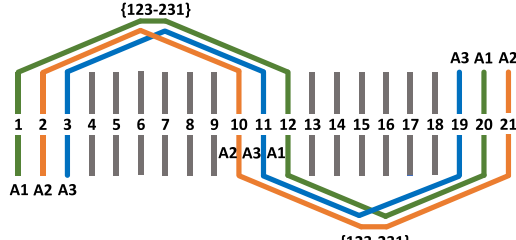
For hairpin windings with multiple parallel paths, achieving impedance symmetry across branches and phases is essential to minimize circulating currents and associated copper losses [85], [86], [87], [88]. In IPM machines, key contributors to impedance imbalance among in-slot conductors include slot leakage flux distortion—often exacerbated by core saturation—and rotor field harmonics [85]. A recent study on a hairpin SPM machine with four parallel branches reported that circulating current losses can contribute up to 12% of total copper losses, highlighting the importance of impedance balance [89]. Transposition helps mitigate this by evenly distributing conductors across parallel paths, ensuring balanced impedance and eliminated circulating currents. In sword-type CHW, transposition is achieved by switching nozzle positions within the same wire set. The wire nozzles associated with a phase can change positions during each winding step, enabling precise conductor placement and effective transposition.

When $q = 2$, there exists one integral pitch movement and $2! - 1 = 1$ transposition movement, named as the transposition patterns. As q increases, the number of possible transposition types ($N_{ip-type}$) for each wire movement, expressed as $N_{ip-type} = q! - 1$ also significantly rises. These transpositions can be named based on their sequence before and after the movement. For instance, {12-21} represents a pattern used for two parallel wires, where the original second wire is placed before the first, as illustrated in Fig. 14(a). Fig. 14(b) and (c) show samples of transposition patterns under $q = 3$ and 4, represented by {123-231} and {1234-2143}, respectively.

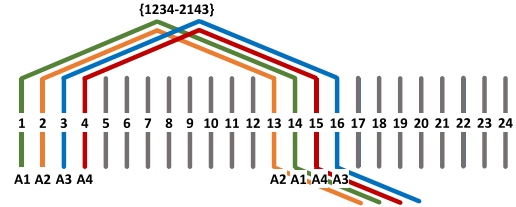
In winding layouts with larger values of q and N_{aa} , employing these transposition patterns is vital for evenly distributing the in-slot positions to each branch. Fig. 15 shows the layout of branch 1-4 of phase A under a CHW stator with $q = 4, p = 4, m = 3, N_L = 8$ and $N_{aa} = 4$: (a) without any transposition,



(a) The transposition pattern under $q = 2$: {12-21}.



(b) One of the transposition patterns under $q = 3$: {123-231}.



(c) One of the transposition patterns under $q = 4$: {1234-2143}

FIGURE 14. Samples of the transposition patterns under $q = 2, 3, 4$ [84].

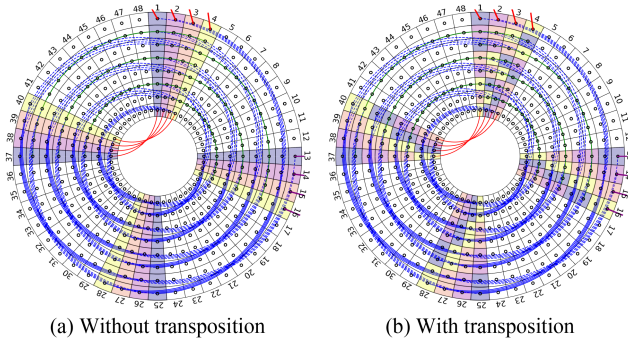


FIGURE 15. Winding layout of phase A in a 48 slots 4 poles 4 parallel branches stator with FBT direction and radial-shift pattern (a) without transposition. (b) With transposition {1234-2341} during layer changing.

and (b) with the transposition of {1234-2341} during the layer changing. It can be noticed that the conductors in (b) are much more extensively distributed across various phasor positions, demonstrating the impact of transposition.

IV. LAYOUT DESIGN RULES AND OBJECTIVES OF CHW

A. SYMMETRY OF CONTINUOUS HAIRPIN WINDING

The manufacturing process of CHW summarized in the previous section imposes the constraints from practical perspective for winding layout design. This section will further integrate

AC winding theory with these constraints to summarize comprehensively the layout design rules for CHW. Overall, the design principle is to achieve “impedance symmetry” between branches of phase winding for the purpose of eliminating circulating current, which follows the same criterion with that of conventional hairpin winding [86]. The impedance of each full branch, embodied as branch resistance and inductance, is the sum of resistance and inductance values of each in-slot conductor within the branch, respectively. For traction/propulsion motors, due to the inherently unbalanced on-load magnetic field distribution in the stator, all the flat conductors that are shifted circumferentially (in different slots) or radially (in different layers) within one phase-pole group correspond to different AC resistance as well as inductance. Hence, these in-slot conductors need to be grouped carefully and then connected electrically by end turns (defined as layout) to form different branches that are with identical impedance [86]. To connect conductors located in all different slots (phasors) and at all different layers, phasor-change jumpers and layer-change jumpers are on one hand necessary for transposition, while on the other hand need to be simplified for practical manufacturing process.

Strong symmetry: With all phase-pole groups equivalently treated, each branch should cover all the different conductor positions radially and circumferentially, by equal number of times. In theory, strong symmetry can eliminate circulating current of CHW despite the manufacturing-caused in-slot positioning deviation [87].

Weak symmetry: With all phase-pole groups equivalently treated, each branch cannot satisfy strong symmetry but still covers all the different conductor layers and phasors, by an equal number of times, respectively [88].

B. BASIC DESIGN RULES ON LAYOUT OF CHW

The following definitions refer to Fig. 10, which shows both the stator winding and the wire mat form of CHW, are given as follows:

- 1) n represents the n^{th} in-slot-position of a wire, shown as the number in each in-slot position.
- 2) $S(n)$, $L(n)$, and $P(n)$ represent the slot index, layer index and phasor index in stator winding form of the n^{th} in-slot-position of a wire, respectively.
- 3) $S_{\text{mat}}(n)$ and $L_{\text{mat}}(n)$ represent the slot and layer index in wire mat form of the n^{th} in-slot-position of a wire, respectively. For major winding area, the relationship between $S_{\text{mat}}(n)/L_{\text{mat}}(n)$ and $S(n)/L(n)$ can be expressed as:

$$S_{\text{mat}}(n) = (L(n) - 1) // 2 \times N_s + S(n)$$

$$L_{\text{mat}}(n) = (L(n) - 1) \% 2 + 1$$

$$P(n) = S(n) // mq + 1 \quad (1)$$

where “//” represents the floor division operator, which returns the largest integer less than or equal to the result of the division, and “%” represents the modulo operator, which returns the remainder after division.

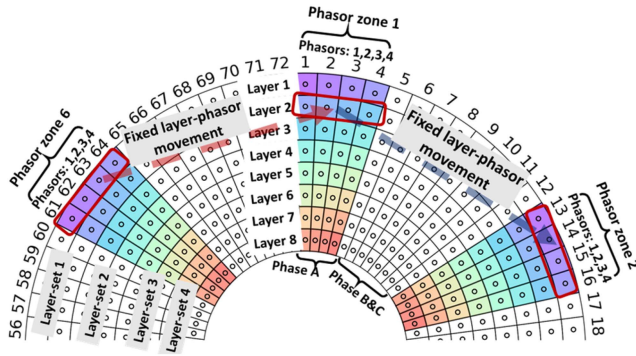


FIGURE 16. Definition of winding zone, layer set, and layer-phasor connection in the stator winding form.

When radial shift exists and the conductors are located at the radial shift area (RSA) where $S(n)$ in $[1, mq]$ and $L(n)$ in $[2, N_L - 1]$, the expression of $L_{mat}(n)$ would be:

$$L_{mat}(n) = L(n)\%2 + 1 \quad (2)$$

- 4) *Phasor zone*: phasor zone is defined as a group of slot positions under the same phase and pole region, which can be seen in Fig. 16, p phasor zones of phase A are evenly distributed including S1-S4, S13-S16, ..., and S61-S64.
- 5) *Layer set*: In corresponding to the 2-layer structure of the wire mat, the term ‘layer-set’ is used to represent each two layers in the stator winding form. For instance, layer 1 and 2 form the first layer set $LS1$, and layer 3 and 4 form the second layer set $LS2$.

In [17], the basic configuration principles/rules are introduced for regular hairpin windings. These rules can be classified into three rules:

Rule#1: The classical AC winding theory should be followed during the settings of coil pitch, phase belt, and winding distribution. For example, the slot number should be multiples of phase number; the conductors of phases under the same pole phase belt should have the same current flow direction; all the phases should have the same coil turns.

Rule#2: The strong symmetry mentioned above needs to be achieved to prevent circulating current, which requires each parallel branch to occupy all the different conductor positions with equal number of times.

Rule#3: For simplification of manufacturing, the winding layout should be as uniform as possible, which requires a limited number of transpositions.

Apart from the above layout rules of hairpin, additional constraints for the CHW due to its unique manufacturing and assembly process should be stated. The constraints from the CHW manufacturing process can then be translated into the following rules.

Rule#4: The first in-slot position of all the wires can only be placed at the first set of layers ($LS1$), and the following in-slot position of wires should follow the fixed layer-phasor

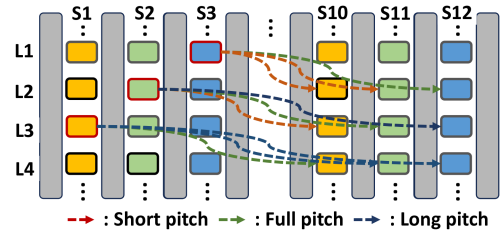


FIGURE 17. An illustration of rule #6 showing all the possible wire movements/end connections when $q = 3$ and $m = 3$.

movement. Fig. 16 shows the fixed movement in CHW with typical sword winding method.

Rule#5: In the stator winding form, only after all the positions on the previous layer set been occupied can any positions on the next layer set be occupied by the conductors. This rule explains the source of the radial shift effect during the transformation from wire mat form to the stator winding form, as shown in Fig. 10. For the regular sword winding method, a uniform layer change (LC) in the wire mat form is required for every transient from conductor to conductor.

$$LC_{mat}(n, n+1) = L_{mat}(n+1) - L_{mat}(n) = \pm 1 \quad (3)$$

For the separate winding sword shown in Fig. 13(b) [33], or similar methods which can reverse the winding direction of the wire mat during the RSA, an additional rule for the layer change $LC_{mat}(n, n+1)$ in wire mat form is that it should remain 0 whenever entering or leaving the radial shift area (RSA).

Rule#6: All the wires should be moved along the same direction from the first layer set to the last layer set.

$$S_{mat}(n+1) > S_{mat}(n) \quad n \in [1, N_{cond/wire}] \quad (4)$$

Particularly, the wire can only move to the q in-slot positions under the next layer-phasor region. Fig. 17 illustrates all the possible wire movements from phasor zone 1 to phasor zone 2.

C. DESIGN OBJECTIVES OF CHW

Due to the unique manufacturing and assembly processes, the design of CHW is less flexible compared to conventional hairpin: 1. The selection of the parallel branches is limited by the wire numbers. 2. The layer jumps of wires are fully determined by the winding pattern. 3. Layer asymmetry due to the radial shift feature. These constraints and features must be considered during the design stage. Some key objectives during the layout design phase are summarized as follows:

- 1) **Cost-effective and feasible manufacturing methods:** Suitable manufacturing processes should be selected based on the winding parameters and the application. For example, choosing the appropriate winding direction based on the required number of parallel branches can save unnecessary welding. Deciding whether to adopt a non-radial-shift pattern based on the motor's

operating conditions and efficiency requirements can optimize the motor's performance and cost.

- 2) Symmetry between parallel branches: Reasonable and necessary slot transpositions should be used to achieve slot symmetry in parallel branches. Corresponding wires that exchange layer positions during the radial-shift region should be series-connected (if possible) to eliminate layer asymmetry. The ideal winding design should achieve strong symmetry to eliminate circulating current losses, but cost and manufacturing consistency should also be considered.
- 3) Minimize irregular slot transpositions: Irregular slot transpositions usually increase wire stress and the end height of the wire mat, causing manufacturing complexity and inconsistency. While achieving symmetry design, irregular slot transpositions should be minimized and, if necessary, placed in different slots to reduce the impact of the increased wire mat height on the rolling up and expansion process.

V. CURRENT AND FUTURE TRENDS IN CONTINUOUS HAIRPIN WINDING DEVELOPMENT

Continuous hairpin winding holds great promise as a mainstream winding technology for the next-generation sustainable transportation, owing to its inherent advantages over regular hairpin winding in loss and efficiency, welding-less feature, and material savings. This potential has spurred recent research on CHW, particularly focusing on the winding process and manufacturability, high-frequency AC losses suppression, and the development of efficient stator cooling. On the other hand, CHW's unique manufacturing process also leads to issues and several challenges, which includes the high bending and assembly forces on wires, complex wire transpositions for strong symmetry, and limited flexibility in wire/winding/stator dimensions and configurations. To accelerate the adoption of CHW as the leading winding process technology for large-scale automotive traction motors, it is essential to effectively address these challenges.

A. WINDING/ASSEMBLY FORCES ON WIRES

CHW is a preformed winding where the coil is axially closed before the assembly [90]. Therefore, it is unable to apply the axially insertion method but instead needs radial insertion [91], which normally refers to the expansion process shown in Fig. 8. This expansion requires specialized machinery and significant force to spread the wire to be squeezed into the stator slots. The bottom surface of conductors is compressed by this high expansion force, leading to deformation and displacement. This could reduce slot fill factor and increase the risk of insulation failure.

To mitigate the radial and tangential forces on winding elements, the company GROB introduced a method in [92] employing support elements to reinforce the winding elements and absorb forces during insertion. GROB then proposed another insertion method in [93] by using centrifugal force. As illustrated in Fig. 18, through the centrifugal force generated

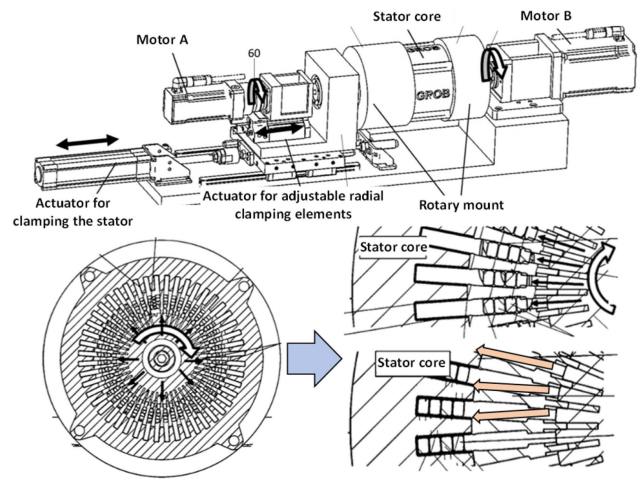


FIGURE 18. CHW assembly using centrifugal force, adapted from [93].

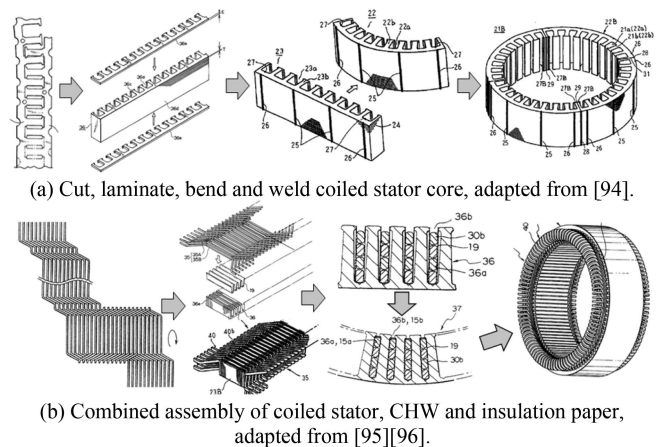
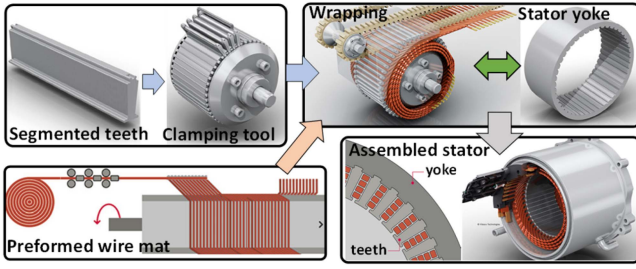


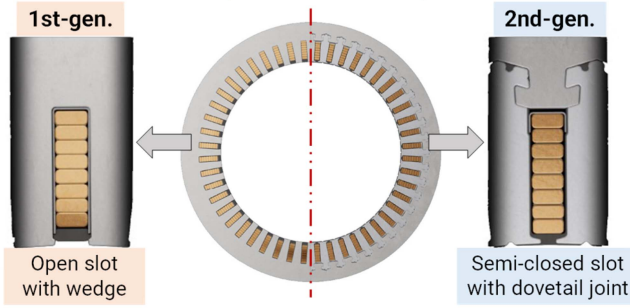
FIGURE 19. (a) Manufacturing process of coiled stator core. (b) Assembly process of coiled stator and continuous hairpin winding [94], [48], [96].

from the rotation of two motors, the winding element is transferred from the tool grooves into the stator slots under the guide of adjustable clamping tools. Compared to the pull-in force, the advantage of centrifugal force lies in its uniform application across all conductors. This significantly reduces the force and damage risk on the innermost conductors and enables slot insulation papers to be clamped more reliably and evenly.

Instead of expanding the wire, deformable or assembled stator core can largely reduce wire deformation. Mitsubishi proposed a processing method for a coiled stator core [94], as shown in Fig. 19(a). Long strip-shaped slotted stator laminations are stacked to form a multi-segment rectangular core, then assembled into a ring-shaped stator through bending and welding. This process can significantly reduce the raw material cost of stator core. Mitsubishi further combined this type of stator with CHW through an integrated bending assembly concept [48], [96], as shown in Fig. 19(b). The long strip-shaped stator core, insulation paper, and winding mat are



(a) Vitesco's manufacturing process of CHW with segmented stator [66]
Photo courtesy of Vitesco Technologies, GmbH.



(b) Schaeffler's 2nd generation CHW with segmented stator [98]
Photo courtesy of Schaeffler GmbH.

FIGURE 20. Vitesco's and schaeffler technologies on the continuous hairpin winding with segmented stator structure [66], [98].

coiled together through bending. This method can improve the consistency of conductor positions in the slot and realize flexibility on slot opening to reduce cogging torque and field harmonics [97].

Segmented stator core is another approach, initially used in [51]. The stator core is divided circumferentially, and its open-slot structure allows the segments to be radially inserted to the winding. Vitesco have developed another segmented stator approach [66], where the teeth are detached from the yoke, allowing for the use of a semi-closed slot structure, as shown in Fig. 20(a). The preformed wire mat is directly wrapped onto the clamping tool that holds the stator teeth, then axially forced into the stator yoke and fixed with dovetail joints after the clamping tool is removed. Schaeffler GmbH is also advancing CHW with segmented stator, as shown in Fig. 20(b) [98]. Segmented design reduces the required joining tolerance and precompression forces of the winding, thus can achieve higher slot fill factor [98]. However, low tolerances between stator iron segments are required to avoid degrading machine performances [99].

B. FLEXIBILITY ON WINDING CONFIGURATIONS

Adopting windings with a high number of slot conductors and parallel branches is a viable solution for high-speed high-power-density machines under limited DC-bus voltage [100]. The number of parallel branches in CHW is usually constrained by the number of inlet wires per phase $N_{\text{wire/phase}}$. When accounting for the strong symmetry possibility, the maximum feasible number of parallel branches per phase can

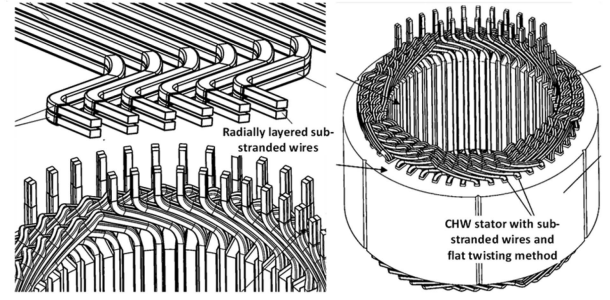


FIGURE 21. Sub-stranded CHW, reproduced from [101].

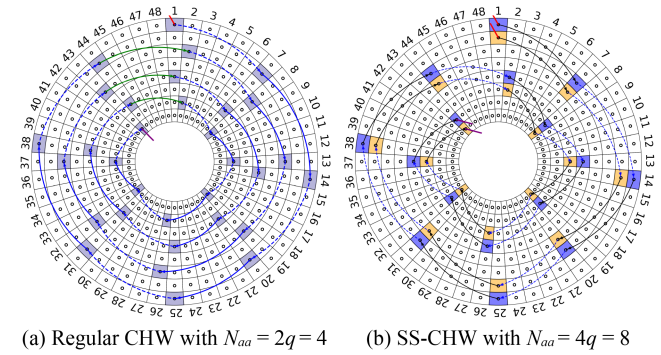


FIGURE 22. The winding layouts of the regular CHW and corresponding sub-stranded CHW under flat twisting method. $p = 8$, $q = 2$, $m = 3$, $N_L = 8$.

be calculated as:

$$N_{aa-\max} = \min(N_{\text{wire/phase}}, p) \quad (5)$$

In conventional CHW for automotive applications, wires per phase $N_{\text{wire/phase}}$ is normally less than the pole number p for the automotive application. To increase the inlet wires of CHW, one approach is to increase the number of interwoven wires using a sub-stranded structure. Fig. 21 illustrates GROB's patents on sub-stranded continuous hairpin winding (SS-CHW). In this design, two flat wires are layered radially and then bent together using flat twisting devices to form the wire mat. This configuration effectively doubles the number of parallel inlet wires from $2q$ to $4q$, also doubling the layer change of wires during each movement.

Fig. 22(a) displays the layout of one branch in regular CHW with $p = 8$, $q = 2$, $m = 3$, $N_{aa} = 4$ and $N_L = 8$. Fig. 22(b) layout of corresponding sub-branches in a 2-sub-stranded flat twisting CHW with $N_{aa} = 8$. Unlike the corresponding CHW, each sub-branch in SS-CHW only occupies half of the layer positions, failing to maintain layer symmetry. To suppress circulating current, these sub-stranded wires need to be preformed to achieve continuous layer transposition.

Phase shifting is recognized as an effective strategy in full pitch hairpin winding traction motors to achieve an equivalent short pitch to reduce harmonics of armature reaction magnetomotive force (MMF) [17]. This feature can also be implemented in CHW through appropriately setting the wire

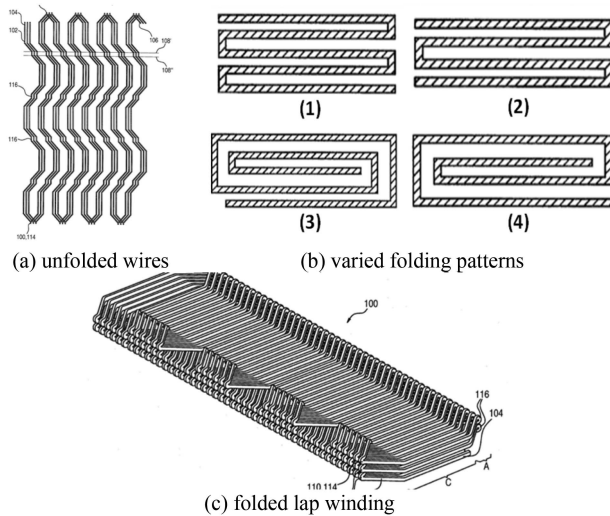


FIGURE 23. Folded continuous lap winding, reproduced from [103].

winding movements for a long (or short) pitch transposition at specific layer change positions [100], [102]. However, this approach comes with more complex wire nozzle movements.

In typical CHW, the layer position of wire is restricted by the winding direction and pattern. Varied folding processes have been identified as a solution to ease these layer position constraints. Robert Bosch GmbH first proposed the organized folding assembly of continuous lap winding [103] (Fig. 23), which was later applied to wave winding [104]. By bending the winding segments towards each other along a folding line in different sequences, the layer position of wires can be adjusted to an optimal design with minimal effort. The layer change pattern can be modified by altering the folding direction and sequence of each sub-wire mat. This approach effectively suppresses the typical radial shift feature and increases the flexibility of layout design [105], [106], [107].

C. AC LOSSES IN HIGH-SPEED APPLICATIONS

In high-speed inverter-fed AC drive systems, higher fundamental frequencies and current harmonics result in uneven current distribution and increased AC copper loss due to both skin and proximity effects, particularly near the slot openings [108]. Segmented/variable cross section design on the regular hairpin winding is recognized as an effective way to suppress the AC copper loss of in-slot conductors [109], [110]. GROB proposed a feasible manufacturing process to align the wires with variable cross sections into a continuous hairpin winding form, allowing CHW to have two or more conductor cross sections [101].

Compared to solid flat copper, using preformed wires with multi-strand conductors can further reduce eddy current loss in these high-frequency applications [111]. To balance the impedances and minimize circulating currents among these conductors, the sub-conductors need to be continuously transposed along the wire's length [112]. These preformed wires

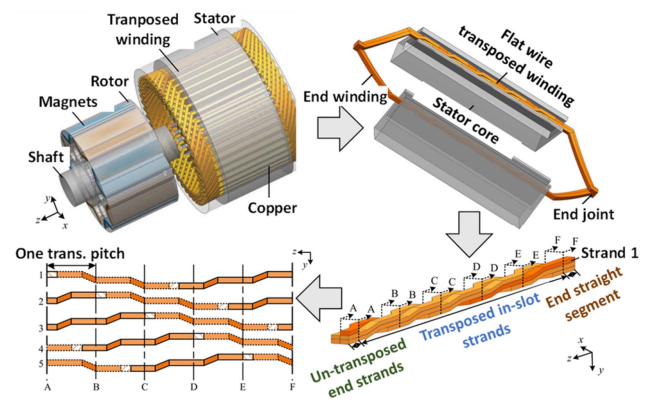


FIGURE 24. Flat wire transposed winding with fully transposed in-slot strands and straight un-transposed end strands, adapted from [119].

include various types, such as automated trickle strands with improved positioning [113], profiled Litz wires [114], transposed Roebel bars [115], flat wires with taped strands [116], additive-manufactured 3D bars with continuous transposition [117], and transposed flat wire with complete transposed in-slot strands and un-transposed end strands [118], [119], as shown in Fig. 24.

Preformed-wire windings share similar structural and assembly characteristics with CHW. For instance, they both require open slots for radial insertion. Compared to I-pin and hairpin, they both offer higher flexibility on turns per slot without welding. They both provide more precise conductor positioning, resulting in higher slot fill factors and reduced AC losses than round wires [120]. Consequently, it is feasible to combine the design and manufacturing methods of CHW with such preformed wires to create high performance winding structures. This integration can significantly enhance the potential of existing CHW designs, particularly in high-frequency applications. However, additional considerations and studies are necessary to evaluate the mechanical, electrical, and thermal characteristics of preformed wires compared to solid wires after bending and assembly process, therefore assess their feasibility for large-scale production [121].

Recently, researchers from RWTH Aachen University and Schaeffler GmbH investigated the production process of profiled Litz wire continuous hairpin winding (PLW-CHW) and its electrical properties and performances post-winding [122], [123], [124]. Compared to the hairpin stator with similar litz wires presented in [125], the continuous wave winding structure eliminates the complex end connections of pin-shaped taped Litz wires. As illustrated in Fig. 25, the PLW-CHW process begins with the profiling and bonding of Litz wires to create compact rectangular strand bundles [122], prior to the regular CHW process. The case study in [123] reveals that for typical electric drive applications with a 220 mm outer diameter and a maximum speed of 18000 rpm, PLW-CHW exhibits a 4% higher DC resistance yet significantly reduced AC resistance at high speeds, leading to improved drive-cycle

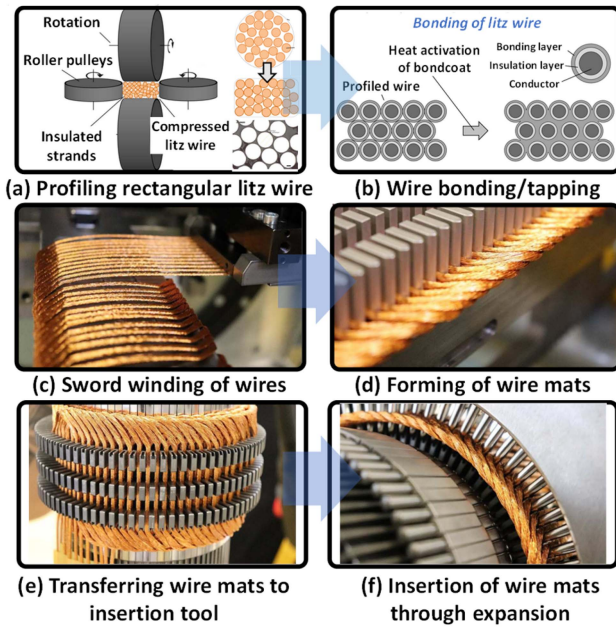


FIGURE 25. Manufacturing process of continuous hairpin winding with preformed flat litz wires, adapted from [122], [123].

efficiency compared to CHW with solid conductors. In [124], the processability of PLW-CHW with different wire insulation and process configurations was investigated regarding to the tested post-process electrical properties, which established processing constraints and design rules for PLW-CHW in relation to the insulation materials, bending angles, and wire dimensions.

D. SIMPLIFICATION ON TRANSPOSITIONS

In previous analyses, minimizing transposition times is regarded as a key design goal for CHW layout, since the transposition increases the manufacturing complexity and inconsistency. For winding configurations that necessitate a higher number of transpositions, recent research and patents have introduced innovative winding methods with specific, periodically transposed patterns along the wire mat. One such method, presented in [84], utilizes a tooling design that automatically alternates the winding's transposition pattern between $\{123-231\}$ and $\{123-123\}$. These automatic transpositions during manufacturing help achieve phasor symmetry, reducing the requirement for introduced irregular connections. Similar strategies are also employed in regular hairpin to reduce required pin shapes [126]. Patent [127] details a manufacturing process where two wires are bent together, creating a uniformly transposed $\{12-21\}$ flat wire mat, as shown in Fig. 26. This method allows for more auto-transposed CHW layouts with even q , such as $\{1234-2143\}$. These winding techniques demonstrate a strategic approach to balancing transpositions with manufacturing efficiency [128].

The regular CHW has low flexibility in achieving strong symmetry through transpositions within the wire. Therefore,

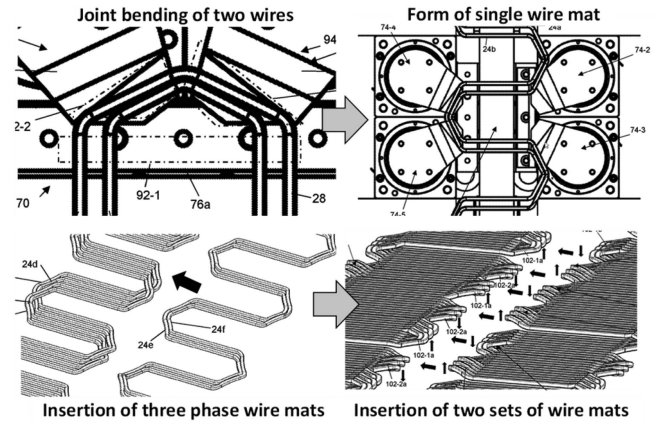


FIGURE 26. Manufacturing and assembly process of CHW with inherent $\{12-21\}$ transpositions, adapted from [127].

segmenting a single continuous wire into multiple serially connected wires helps avoid complex wire transpositions, simplifying the realization of strong symmetry. By assembling and reconnecting the segmented wires, the strong symmetry winding pattern can be achieved without using any interweaving of wires, i.e., transpositions, as illustrated in [129]. Careful design of wire span and connections can achieve a simplified wire scheme and strong symmetry with a minimal number of connections. This pseudo-continuous winding effectively increases the design flexibility and application range of CHW.

E. DIRECT COOLING OF CHW

Efficient thermal management is key to achieving high power density in traction motors. Recent trends in cooling for hairpin winding traction motors have focused on bringing the coolant closer to the winding parts, thus enhancing heat dissipation capabilities [130]. These methods include end-winding oil spray/jet [19], in-slot direct oil cooling [132], and hollow/profiled conductor direct cooling [133].

The superior thermal conductivity of copper compared to insulation and iron materials significantly enhances the efficiency of cooling inside the conductor and substantially boosts continuous power density [134]. Stefan et. al [135] proposed a novel integration of in-conductor direct cooling with a multi-phase low voltage traction motor, which negates the need for insulation coating and allows direct contact between the coolant and conductors. As shown in Fig. 27(a), 48 identical W-shaped coils create a “continuous” 48-phase wave winding. The increased DC resistance caused by the centre opening is effectively compensated by the lower conductor temperature, allowing a constant current density of up to 49.5 A/mm^2 . However, characteristics of hollow wires influenced by different processing methods remain unclear, which need to be systemically assessed through sampling and testing [136].

For conventional 3-phase medium voltage traction motors, it's important to consider the appropriate design of the cooling path. An excessively long path increases the required pressure

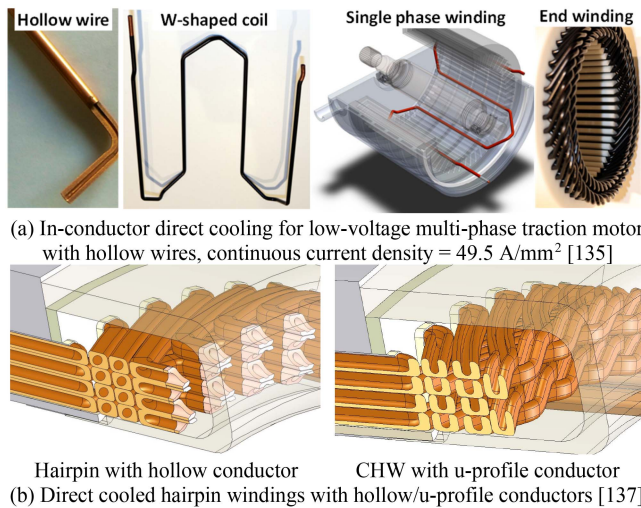


FIGURE 27. In-conductor direct cooling with hollow and profiled conductors in hairpin windings, reproduced from [135], [137].

and reduces coolant flow. For regular hairpin windings, hollow conductors can be used with the ends of the pins serving as the coolant's inlet and outlet, and each pin operates in parallel within the cooling system [137], as shown in Fig. 27(b). When using continuous hairpin winding with much longer single wires, a closed hollow wire could result in an excessively long cooling path. In such cases, using profiled wires, as illustrated in Fig. 27(c), is more applicable [137]. The cooling medium is guided through the grooves on the conductor to flow unidirectionally from one end of the stator to the other.

F. FUTURE DIRECTIONS ON CHW RESEARCH AND DEVELOPMENT

Future directions primarily focus on advancing the Technology Readiness Level (TRL) of existing CHW techniques, particularly emphasis on the two major challenges during winding and assembly processes:

- 1) **The manufacturing-related winding feature**, which leads to weak symmetry and limitation to winding configurations. Innovative design and manufacturing techniques are being developed to tackle the challenges posed by radial shift, particularly in simplifying winding tools and transposition patterns.
- 2) **The high forming and assembly pressure**, which can cause deformation and displacement of windings. Comprehensive testing and evaluations [138] are essential for the advancement and maturation of novel assembly technologies like segmented stators, aimed at reducing costs, enhancing reliability, and improving electromagnetic performance.

Additionally, CHW faces several design constraints inherent to its manufacturing process [61]. These constraints include limitations on the wire aspect ratio and the winding diameter ratio (the ratio between inner and outer diameters), primarily due to the force required for insertion. Moreover,

tools often need to be customized according to the dimensions of the winding and conductors to ensure compatibility in terms of sizes and tolerances, thus restricting the design and modification flexibility of CHW. Many manufacturers are actively working on developing more flexible assembly tools and scalable automated production chains. These advancements aim to accommodate the manufacture of motors in various sizes and designs, saving costs and reducing production cycles.

In alignment with global objectives to achieve net-zero emissions, technological roadmaps for traction motors are increasingly concentrating on the sustainability and environmental aspects of large-scale production [139], [140]. In addition to enhancing power density for material use, factors like maintainability and the recyclability of materials are being increasingly considered, especially in the processing of windings and magnets. Innovations stemming from life cycle assessments of current manufacturing methods [141] are leading to designs that facilitate disassembly, efforts to minimize wet processes and heat treatments in assembly, aimed at boosting material recycling rates. The smaller end winding of CHW leads to a reduced copper usage. Additionally, the weldless nature of CHW facilitates easier disassembly or replacement [61], thus improving sustainability.

The use of new conductor materials, such as Al-Cu alloy [140], will aid in achieving stator lightweighting, cost reduction, and performance enhancement. CHW can be combined with innovative manufacturing processes like additive manufacturing to further integrate or reduce assembly steps related to insulation, connection, and impregnation, thus minimizing carbon emissions, and accelerating production pace. Advancements in materials and manufacturing techniques [141] will significantly enhance the flexibility of CHW, enabling the enhanced-performance or multi-functional designs with functionally graded materials, cross sections, and insulation properties.

VI. CONCLUSION

This paper has presented a comprehensive review of the state-of-the-art continuous hairpin winding (CHW) technology. After highlighting the differences between CHW and other winding types regarding basic features such as AC losses and dependency of welding, the focus of review has been made on manufacturing processes (including alternative implementation routes), layout design principles, and recent research developments. The review has concluded that design flexibility and complexity of CHW are closely linked to the selected manufacturing approaches. Recent advancements in technology building blocks and academic studies are actively addressing key challenges related to CHW, including manufacturing consistency, scalability, reliability, and standardization.

The continued evolution of CHW has been primarily driven by industry-led R&D due to the high capital requirements, with contributions from a diverse range of companies. Meanwhile, academic engagement is growing, particularly in the development of design methodologies and performance

benchmarking. As a viable alternative to conventional hairpin winding in electric transportation systems, significant progress is being made in CHW development both in electromagnetic design and mass production oriented manufacturing technologies.

Along with increasing speed, frequency, and voltage demands of future EV and aerospace propulsion systems being pushed, CHW has been demonstrated with considerable advantages over both conventional hairpin and stranded windings. Its ability to accommodate higher slot and layer numbers allows for smaller conductor to reduce AC losses. The elimination of welding further enhances the insulation reliability, making it a promising winding topology for high voltage system.

Future research will focus on broadening CHW's application range, increasing cost-effectiveness, and further improving its performance metrics. With ongoing progress in materials science, process automation, and system integration, CHW is well-positioned to become a cornerstone technology for the next generation of sustainable, high-performance and high reliability drive systems.

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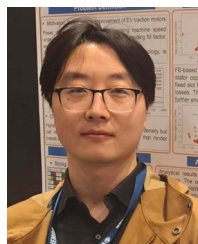
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